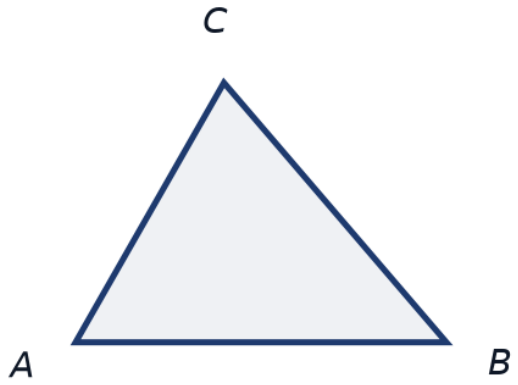
Geometry & Measurement

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

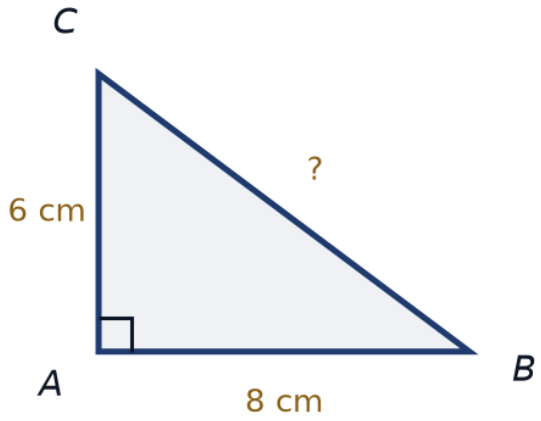
1 **[Foundational]** • **Target: Triangle angle sum**

Find the third angle of the triangle shown, given the two marked angles.



2 **[Foundational]** • **Target: The Pythagorean theorem**

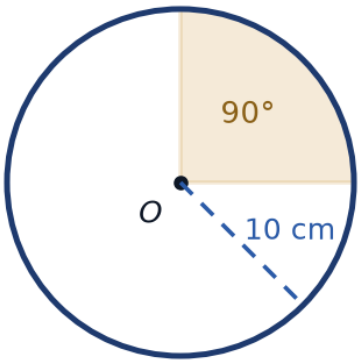
Find the hypotenuse of the right triangle shown.



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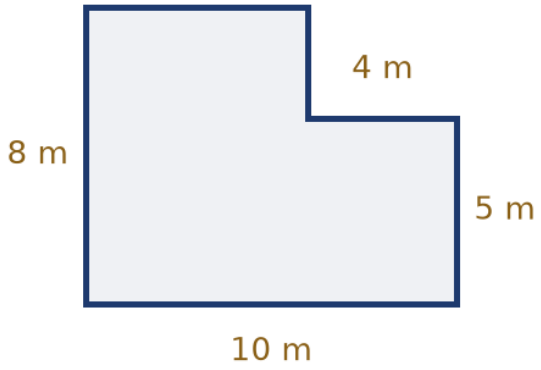
3 [Foundational] • Target: Circles: circumference and area

Find the area of the shaded 90° sector shown, in terms of π.



4 [Foundational] • Target: Composite figures

Find the area of the L-shaped room shown.



5 [Foundational] • Target: Angle facts

Two angles on a straight line measure $3x$ and $2x + 30$. Find x.

6 [Foundational] • Target: Angle facts

Find the angle vertically opposite a 65° angle.

7 [Foundational] • Target: Angle facts

Three angles around a point measure 110°, 95°, and x. Find x.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Angle facts

Two angles are complementary (sum to 90°). One measures 37°. Find the other.

9 [Intermediate] • Target: Parallel lines and transversals

Two parallel lines are cut by a transversal, creating an angle of 70°. Find its corresponding angle.

10 [Intermediate] • Target: Parallel lines and transversals

In the same setup, find the co-interior (allied) angle to a 70° angle.

11 [Intermediate] • Target: Parallel lines and transversals

Find the alternate interior angle to a 55° angle.

12 [Intermediate] • Target: Parallel lines and transversals

Two parallel lines cut by a transversal create an angle of $4x$ and its co-interior angle of $2x + 30$. Find x.

13 [Intermediate] • Target: Triangle angle sum

An isosceles triangle has two equal angles of 72° each. Find the third angle.

14 [Intermediate] • Target: Triangle angle sum

In a right triangle, one of the non-right angles is 35°. Find the other non-right angle.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Triangle angle sum • Cross-Lesson: 1.01 Arithmetic & Number Sense

A triangle has three equal angles. Find each angle, and name this type of triangle.

16 [Challenging] • Target: Angle sums of polygons • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Find the sum of the interior angles of a hexagon (6 sides).

17 [Challenging] • Target: Angle sums of polygons • Cross-Lesson: 3.01 Algebra Essentials

Find the sum of the interior angles of a pentagon (5 sides).

18 [Challenging] • Target: Angle sums of polygons • Cross-Lesson: 5.01 Quantitative Comparisons

A regular polygon has an interior angle sum of $1{,}080^\circ$. How many sides does it have?

19 [Challenging] • Target: The Pythagorean theorem • Cross-Lesson: 6.01 Sequences & Patterns

A right triangle has hypotenuse 13 and one leg 5. Find the other leg.

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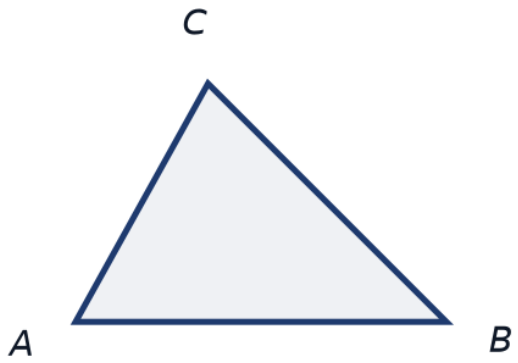
20 [Challenging] • Target: The Pythagorean theorem • Cross-Lesson: 7.01 Data Analysis & Probability

A 10 m ladder leans against a wall with its foot 6 m from the wall's base. How high up the wall does it reach?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Triangle angle sum

Find the third angle of the triangle shown.



a. 116°

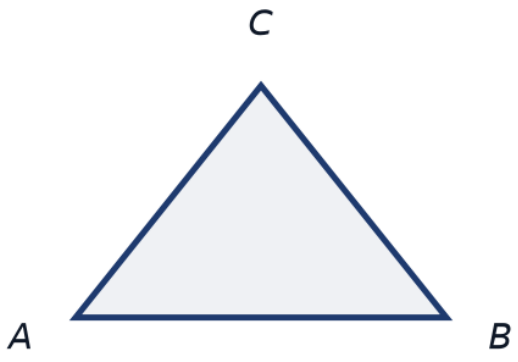
b. 64°

c. 54°

d. 44°

22 [Foundational] • Target: MCQ — Triangle angle sum

The isosceles triangle shown has two equal base angles of 70° each. Find the apex angle.



a. 55°

b. 70°

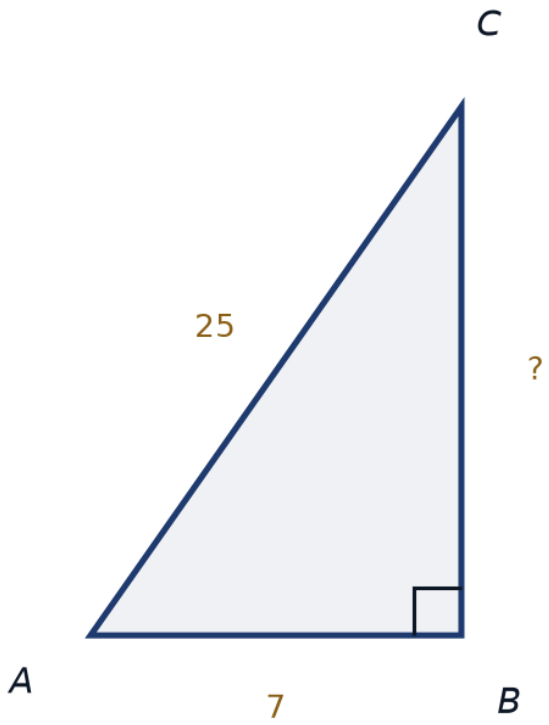
c. 40°

d. 110°

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23 [Foundational] • Target: MCQ — The Pythagorean theorem

Find the missing leg of the right triangle shown.



a. 26

b. 18

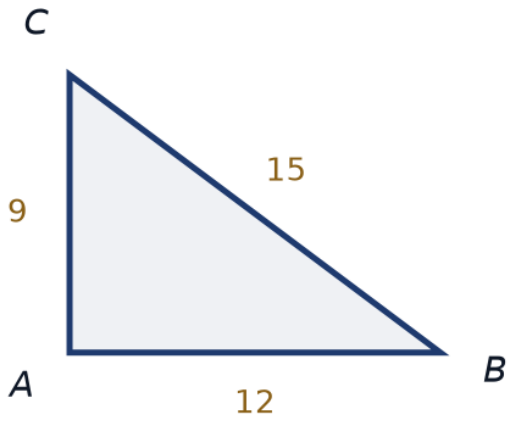
c. 24

d. 20

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24 [Foundational] • Target: MCQ — The Pythagorean theorem

A triangle has sides 9 cm, 12 cm, and 15 cm, as shown. Is it a right triangle?



a. Only if it is also isosceles

b. Yes — a right triangle

c. Cannot be determined

d. No — not a right triangle

25 [Foundational] • Target: MCQ — Perimeter and area of rectangles and squares

Find the perimeter and area of the rectangle shown.

a. $P = 48$ cm, $A = 135$ cm^2

b. $P = 48$ cm, $A = 24$ cm^2

c. $P = 135$ cm, $A = 48$ cm^2

d. $P = 24$ cm, $A = 135$ cm^2

26 [Foundational] • Target: MCQ — Perimeter and area of rectangles and squares

Find the perimeter and area of the square shown.

a. $P = 44$ cm, $A = 110$ cm^2

b. $P = 44$ cm, $A = 121$ cm^2

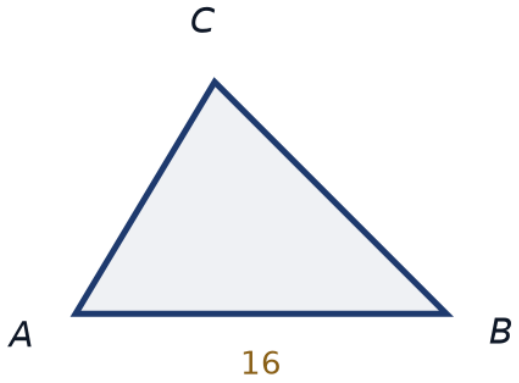
c. $P = 22$ cm, $A = 121$ cm^2

d. $P = 121$ cm, $A = 44$ cm^2

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27 [Foundational] • Target: MCQ — Area of triangles and parallelograms

The triangle shown has base 16 cm and height 10 cm. Find its area.



a. 160

b. 26

c. 80

d. 40

28 [Foundational] • Target: MCQ — Area of triangles and parallelograms

The parallelogram shown has base 18 cm and height 7 cm. Find its area.

a. 252

b. 126

c. 63

d. 25

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Angle facts

Two angles are supplementary. One is three times the other. Find the smaller angle.

a. 60°

b. 30°

c. 135°

d. 45°

30 [Intermediate] • Target: MCQ — Parallel lines and transversals

Two parallel lines are cut by a transversal. One angle is $(3x + 15)^\circ$ and its corresponding angle is $(5x - 25)^\circ$. Find x.

a. 10

b. 8

c. 30

d. 20

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31 [Intermediate] • Target: MCQ — Triangle angle sum

A triangle has three angles in the ratio $2:3:4$. Find the largest angle.

a. 90°

b. 100°

c. 80°

d. 60°

32 [Intermediate] • Target: MCQ — Perimeter and area of rectangles and squares

A square has area $225\ \text{cm}^2$. Find its side length and perimeter.

a. $s = 15$ cm, $P = 30$ cm

b. $s = 225$ cm, $P = 900$ cm

c. $s = 15$ cm, $P = 60$ cm

d. $s = 45$ cm, $P = 60$ cm

33 [Intermediate] • Target: MCQ — Composite figures

A garden is L-shaped: an 18 m by 8 m rectangle with a 6 m by 3 m rectangle removed from one corner. Find its area.

a. 108

b. 126

c. 162

d. 138

34 [Intermediate] • Target: MCQ — Similar figures and scale factor

Two similar triangles have a scale factor of $3:5$. If the smaller triangle's perimeter is 24 cm, find the larger triangle's perimeter.

a. 14.4

b. 72

c. 64

d. 40

35 [Intermediate] • Target: MCQ — Word problems

Tiling costs 38 riyals per square meter. Find the total cost to tile a rectangular room measuring 9 m by 6 m.

a. 1,026 SAR

b. 4,104 SAR

c. 342 SAR

d. 2,052 SAR

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36 [Intermediate] • Target: MCQ — Perimeter and area of rectangles and squares

A right triangle has legs 33 cm and 180 cm. Find the hypotenuse.

a. 183

b. 181

c. 213

d. 185

37 [Intermediate] • Target: MCQ — Area of triangles and parallelograms

A right triangle has legs 20 cm and 48 cm. Find the hypotenuse.

a. 54

b. 52

c. 50

d. 68

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Circles: circumference and area • Cross-Lesson: 5.01 Quantitative Comparisons

A right triangle has legs 27 cm and 120 cm. Find the hypotenuse.

a. 121

b. 125

c. 123

d. 147

39 [Challenging] • Target: MCQ — Composite figures • Cross-Lesson: 6.01 Sequences & Patterns

A rectangle has length 9 cm and width 8 cm. Find its perimeter and area.

a. $34, 78$

b. $38, 72$

c. $34, 72$

d. $72, 34$

40 [Challenging] • Target: MCQ — Volume and surface area • Cross-Lesson: 7.01 Data Analysis & Probability

A rectangle has length 6 cm and width 4 cm. Find its perimeter and area.

a. $24, 24$

b. $20, 24$

c. $20, 30$

d. $24, 20$

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41 [Challenging] • Target: MCQ — Coordinate geometry: distance and midpoint • Cross-Lesson: 8.01 Word Problems & Reasoning

A rectangle has length 12 cm and width 6 cm. Find its perimeter and area.

a. $36, 72$

b. $36, 78$

c. $72, 36$

d. $40, 72$

42 [Challenging] • Target: MCQ — Similar figures and scale factor • Cross-Lesson: 1.01 Arithmetic & Number Sense

A rectangle has length 21 cm and width 14 cm. Find its perimeter and area.

a. $294, 70$

b. $70, 294$

c. $70, 300$

d. $74, 294$

43 [Challenging] • Target: MCQ — Word problems • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

An isosceles triangle has two equal angles of 64° each. Find the third angle.

a. 52

b. 46

c. 64

d. 58

44 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Algebra Essentials

Find the sum of the interior angles of an octagon (8 sides).

a. 1080

b. 1440

c. 1260

d. 900

45 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Quantitative Comparisons

A circle has radius 35 cm. Find its area, using $\pi = \frac{22}{7}$.

a. 3864

b. 220

c. 3850

d. 3836

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